Mohammad Asif Hasan

B.Sc. in Electrical and Electronic Engineering, BUET • Dhaka, Bangladesh

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Research Interests

Robust and interpretable deep learning for medical image analysis (breast ultrasound and mammography): segmentation and diagnosis models whose failures are measurable in clinical terms, multimodal fusion of images with clinical data, and evaluation under distribution shift. Secondary interest in efficient visual perception for edge and robotic systems.

Education

Bangladesh University of Engineering and Technology (BUET) Jan 2022 – Jul 2026
B.Sc. in Electrical and Electronic Engineering Dhaka, Bangladesh

- Thesis: *Toward Clinically Interpretable Breast Cancer Imaging*, joint with Md Nazmus Sakib Tushar; supervised by Prof. Lutfa Akter.
- Relevant coursework: Introduction to Medical Imaging, Digital Signal Processing, Artificial Intelligence and Machine Learning, Introduction to Robotics, Linear Algebra, Probability and Statistics.

Notre Dame College, Dhaka 2018 – 2020
Higher Secondary Certificate (Science), GPA 5.00 / 5.00

Nasirabad Govt. High School, Chattogram 2016 – 2018
Secondary School Certificate (Science), GPA 5.00 / 5.00

Publications

Peer-reviewed conference papers

[1] **M. A. Hasan**, N. Ahmed, and M. Ahmed, "Real-time Breast Cancer Classification Employing GMM-HMRF Segmentation and PCA-Optimized Machine Learning Algorithms," in *Proc. 28th Int. Conf. on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025. IEEE Xplore | Code

[2] **M. A. Hasan**, M. S. Hasan, A. Hamim, N. Ahmed, S. Imam, and T. Ahmed, "Multimodal Pipeline for Sentiment Mining via MAG-BERT, LoRA-T5 Generated Suggestions with SHAP Explanations for Curriculum Optimization," in *Proc. 28th Int. Conf. on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025. IEEE Xplore | Code

[3] F. Labiba, **M. A. Hasan**, S. Sarkar, A. Shahriar, N. Tasnim, and S. A. Fattah, "Real-Time Multi-Modal Drone and Bird Tracking Using Modality-Aware Adaptation and Physics-Aware Filtering," in *Proc. IEEE Int. Women in Engineering (WIE) Conf. on Electrical and Computer Engineering (WIECON-ECE)*, Cox's Bazar, Bangladesh, 2025. IEEE Xplore

[4] A. S. M. Asaduzzaman, M. M. Hasan, M. K. Jahan, M. A. A. Chowdhury, **M. A. Hasan**, and M. Arman, "An Improved Optimization Approach for Secure User Authentication in Cyber Physical Systems," in *Proc. Int. Conf. on Computing Technologies*, Bengaluru, India, 2025.

Preprint

[5] S. Sarkar, **M. A. Hasan**, K. A. Shahriar, F. Labiba, N. Tasnim, and S. A. Fattah, "EGD-YOLO: A Lightweight Multimodal Framework for Robust Drone-Bird Discrimination via Ghost-Enhanced YOLOv8n and EMA Attention under Adverse Condition," arXiv:2510.10765, 2025. arXiv

Manuscripts in preparation

[6] **M. A. Hasan**, et al., "HybridSwinUNet: A Boundary-Aware, Style-Randomized Attention Network for Robust Breast Ultrasound Lesion Segmentation Under Rigorous Cross-Validation," journal manuscript in preparation.

[7] S. Sarkar, R. Mahmud, **M. A. Hasan**, and M. M. Islam, "Memes Move Faster Than Models: Script, Prior, Shortcut and Temporal Shift in Multimodal Bangla Political Meme Detection," manuscript in preparation.

[8] BUET VIP Cup team, including **M. A. Hasan** (author order to be finalised), "Weighted Ensemble Fusion of Heterogeneous RGB–IR Detectors for Small-Object Drone–Bird Detection: Architectural Ablation and the Accuracy–Latency Trade-off," manuscript in preparation.

Research Experience

Undergraduate Thesis, BUET Jan 2026 – Jun 2026
Supervisor: Prof. Lutfa Akter, Dept. of EEE; joint with Md Nazmus Sakib Tushar Dhaka, Bangladesh
Toward Clinically Interpretable Breast Cancer Imaging: A Boundary-Aware Hybrid Framework for Ultrasound Lesion Segmentation and Multimodal Mammographic Classification with BI-RADS-Aware Fusion

- **Mammography:** Lesion-Aware Attention U-Net for pectoral muscle removal, with a new metric, the Lesion Preservation Rate (LPR). DSC 0.897; LPR 97.7% vs. 81.1% for the best classical baseline ($p < 0.001$).

- **Breast ultrasound:** boundary-aware hybrid attention network. Dice 77.28 / 82.17 / 88.61% on BUSI / UDIAT / BUS-BRA under stratified cross-validation; significantly above the strongest of five baselines on BUSI and UDIAT, comparable on BUS-BRA, with the lowest fold variance on all three.
- **BI-RADS classification:** BOCAF, a bidirectional ordinal cross-attention model fusing mammograms with clinical metadata. QWK 0.793, macro AUC 0.841 on VinDr-Mammo.
- Tools: PyTorch, 3D Slicer, SimpleITK • Data: CBIS-DDSM, VinDr-Mammo, BUSI, UDIAT, BUS-BRA

Undergraduate Researcher, Prof. Shaikh Anowarul Fattah's group, BUET 2025
Multimodal (RGB–infrared) drone detection and tracking; IEEE VIP Cup 2025 team Dhaka, Bangladesh

- Built the Ghost and EMA modules and the RGB–IR data pipeline for EGD-YOLO [5]; ran the experiments and ablations and co-wrote the paper.
- Built the detection model for the real-time multimodal tracker [3]; ran the tracking experiments and co-wrote the paper.

Research Collaborator, Dept. of EEE, BUET 2026 – present
With S. Sarkar and colleagues Dhaka, Bangladesh

- Temporal distribution shift in multimodal Bangla political meme detection [7]; weighted fusion of RGB and thermal detectors [8].

Honors and Awards

- **IEEE VIP Cup 2025 – Second Runner-Up (third place)**, Team EyeQ; final at ICIP 2025, Anchorage, Alaska. Infrared–visual drone detection, tracking and payload identification; I built the team's YOLOv8 detector and payload-recognition model. 2025
- **IEEE ICIP 2025 Travel Grant**, IEEE Signal Processing Society, for the VIP Cup final. 2025
- **Finalist**, Datathon, CUET CSE Fest 2025 (Team Breakpoint): political meme classification with PaddleOCR. 2025
- **Top 30**, NNO Content Writing Hunt Programme 2021
- **2nd Position**, Divisional Science Fair, Chattogram Division 2017

Selected Projects

- **HIVE-R**, bio-inspired swarm drones for human detection in GPS-denied areas (GitHub): deployed a 6.2 MB YOLOv8n on a Raspberry Pi 4 (86.9% mAP under smoke and dust, 185 ms per frame); EKF-fused monocular VO and IMU for 6-DoF navigation; 3D target coordinates relayed over 915 MHz MAVLink in under 500 ms.
- **AstroVeritas**, NASA Space Apps Challenge 2025, team project (GitHub): exoplanet candidate vetting from TESS light curves with synthetic transit injection and LightGBM.

Work Experience

BRIM AI – AI System Development Engineer May 2026 – Sep 2026
Real-time video analytics for security surveillance Dhaka, Bangladesh

- Developed a computer-vision agent for room security surveillance with real-time threat detection; optimised the model for on-device inference and deployed it in a user-facing mobile app.

Technical Skills

Deep learning	PyTorch, TensorFlow, Keras, scikit-learn, YOLOv8, NumPy, Pandas, OpenCV, Matplotlib
Medical imaging	3D Slicer, SimpleITK, scikit-image
Languages	Python, C, C++, MATLAB, Verilog
Systems & tools	Raspberry Pi, Arduino, MAVLink, Git, Docker, GCP, Linux, LaTeX

Leadership and Service

- Vice President, BUET Robotics Society, 10th Executive Committee (Mar 2025 – Jul 2026); led an introductory robotics and Arduino workshop (Jan 2025, post).
- Webmaster, IEEE EMBS Student Branch Chapter, BUET (Jun 2025 – present); IEEE Student Member since 2021.
- Content Lead, Newspaper Olympiad (Jul 2021 – May 2022).

References

Prof. Lutfa Akter (thesis supervisor)
Professor, Dept. of EEE, BUET
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Prof. Shaikh Anowarul Fattah (research advisor)
Professor, Dept. of EEE, BUET
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